KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

The Semester GPA and CGPA Calculation System

Submitted by

Karthick. R

25BCS225

Under the Guidance of

Mr.Dharaneesh

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

Edit with WPS Office

I hereby declare that the project entitled“vehicle Rental Booking and Payment Management System” submitted in partial fulfillment of the requirements for the course Java programming is a **record of my original work** carried out under the guidance of **Mr.Dharaneesh**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

Edit with WPS Office

CERTIFICATE

This is to certify that the project entitled **“ vehicle Rental Booking and Payment Management System** is a **bona fide work** carried out by karthick RB.Sc(25-28) **Computer science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Edit with WPS Office

Guide:
(Technical Trainer, KGCAS)
KGCAS)

Head of the Department
(Technical Head,

_____ _____

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	

Edit with WPS Office

7.2	Conclusion	
8	**References**	

ABSTRACT

The **Vehicle Rental Booking and Payment Management System** is a Java-based application developed to simplify and automate the process of vehicle rental booking and payment handling. In many rental services, booking and payment details are maintained manually, which may lead to errors, confusion, and delays in service.

This system allows users to view available vehicles, choose a vehicle for rent, enter rental details, calculate the rental amount, and process the payment. The system is designed using Core Java concepts such as classes, objects, methods, loops, arrays, and conditional statements.

The application provides a simple and user-friendly way to manage vehicle bookings and payments efficiently. It helps reduce manual effort, avoids calculation mistakes, and improves the speed of service. The system can be useful for small vehicle rental businesses such as bike rental, car rental, and van rental services.

The application provides a simple and user-friendly way to manage vehicle bookings and payments efficiently. It helps reduce manual effort, avoids calculation mistakes, and improves the speed of service. The system can be useful for small vehicle rental businesses such as bike rental, car rental, and van rental services.

The main objective of this project is to provide an efficient and accurate rental management solution while demonstrating the practical application of Java programming concepts in a real-world business environment.


Edit with WPS Office

1. INTRODUCTION

Vehicle rental services have become increasingly popular because they offer convenience and flexibility to customers who need transportation for a short period. Rental agencies provide various types of vehicles such as bikes, cars, vans, and scooters based on customer needs.

In many small-scale rental businesses, booking and payment details are still handled manually using paper records or calculators. This manual process can create problems such as booking confusion, incorrect bill calculation, data loss, and delays in customer service. It also becomes difficult to keep track of vehicle availability and payment status.

To overcome these issues, the **Vehicle Rental Booking and Payment Management System** is developed as a computerized solution. This system helps staff or users book vehicles easily, calculate rental charges automatically, and manage payment details in an organized way.

The system allows users to select a vehicle, enter customer and rental information, calculate the cost based on rental duration, and display the final payment amount. It reduces manual work and makes the rental process faster and more accurate.

This project is developed using Core Java programming and demonstrates how programming concepts can be used to solve real-life problems in business applications.

1.1 Objective

The main objectives of the Vehicle Rental Booking and Payment Management System are:

- To develop a Java-based rental booking system

Edit with WPS Office

- To automate vehicle booking and payment calculation
- To reduce manual errors in rental management
- To provide quick and accurate billing
- To maintain clear rental and payment details
- To demonstrate the practical use of Core Java concepts

1.2 Overview

The Vehicle Rental Booking and Payment Management System is a menu-driven application developed using Java. It helps manage vehicle rentals by allowing users to select available vehicles, enter rental duration, and generate payment details automatically.

The system displays different types of vehicles and their rental charges. The user selects a vehicle, enters the number of rental days or hours, and the system calculates the total amount. It can also display booking confirmation and payment information.

This project is useful for rental businesses because it improves efficiency, accuracy, and customer service.

1.3 Features

The main features of the system are:

- Vehicle availability display
- Vehicle booking process
- Rental duration entry
- Automatic charge calculation
- Payment management
- Booking summary generation
- Simple and user-friendly interface
- Fast and accurate processing


Edit with WPS Office

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied carefully and the system requirements are identified clearly. It helps in understanding the existing issues and finding the best computerized solution.

For the Vehicle Rental Booking and Payment Management System, the analysis focuses on the problems involved in manual booking and payment processing. It also identifies the required inputs, outputs, and functionality of the proposed system.

By performing proper analysis, the developed system becomes more reliable, easy to use, and efficient in handling rental operations.

2.1 Problem Definition

In many rental businesses, vehicle booking and payment handling are done manually. This creates several problems such as:

- Errors in rental amount calculation
- Difficulty in checking vehicle availability
- Time-consuming booking process
- Improper maintenance of customer records
- Confusion in payment status
- Delays in generating rental bills

Edit with WPS Office

Manual systems increase the workload of staff and may affect customer satisfaction. Therefore, there is a need for an automated system that can efficiently manage bookings and payments.

2.2 Project Overview

The Vehicle Rental Booking and Payment Management System is a Java-based application that automates rental booking and payment calculation.

The system allows users to:

- View available vehicles
- Choose a vehicle for rent
- Enter rental duration
- Calculate rental charges
- Display final booking and payment details

This project is developed using Core Java programming concepts and can be implemented as a console-based application.

2.3 Scopes and Limitations

Scope

The scope of the system includes:

- Managing vehicle rental bookings
- Calculating rental charges
- Displaying payment details
- Providing booking summaries
- Improving rental management efficiency

The system can be used in:

- Car rental services
- Bike rental centers
- Scooter rental businesses
- Travel vehicle rental agencies

Limitations

The limitations of the system are:

- It is a basic console-based application


Edit with WPS Office

- It does not store data permanently in a database
- It does not include online payment gateway integration
- It is suitable mainly for small-scale rental management

3. SYSTEM DESIGN

System design is the phase where the structure of the system is planned before implementation. It describes how different parts of the application work together to achieve the required result.

For the Vehicle Rental Booking and Payment Management System, the design is divided into modules such as vehicle display, booking, calculation, and payment management. This modular structure makes the system easier to understand and develop.

Proper system design ensures smooth functioning, clarity, and better performance of the application.

3.1 Modules

1. Vehicle Display Module

This module displays the list of available vehicles along with their rental charges.

Example:

Edit with WPS Office

1. Bike – ₹500 per day
2. Car – ₹1500 per day
3. Scooter – ₹400 per day
4. Van – ₹2500 per day

This helps the user choose the required vehicle easily.

2. **Booking Module**

This module allows the user to select the vehicle and enter booking details such as:

- Customer name
- Vehicle type
- Number of rental days

This module stores the booking information temporarily for further processing.

3. **Rental Calculation Module**

This module calculates the rental cost based on the selected vehicle and number of rental days.

Formula:

Rental Amount = Daily Rate × Number of Days

If needed, additional charges can also be added.

4. **Payment Management Module**

This module manages payment details by:

- Displaying total rental amount
- Accepting payment amount
- Checking whether payment is full or partial
- Showing payment balance if any

5. **Booking Summary Module**

This module generates the final summary of the booking, including:

- Customer name
- Vehicle selected
- Rental duration
- Total amount
- Payment status

Edit with WPS Office

6. Exit Module

This module safely exits the application after completing the booking and payment proces

4. METHODOLOGY

Methodology refers to the systematic approach or process followed to develop a software project. It describes the sequence of steps taken from understanding the problem to designing, implementing, testing, and documenting the system. A well-defined methodology ensures that the project is developed in an organized and efficient manner.

In software development, methodology helps in dividing the entire project into smaller stages such as problem analysis, system design, program development, testing, and documentation. This structured approach makes it easier to manage the project and ensures that all requirements are properly addressed.

4.1. Problem Analysis

Edit with WPS Office

In this phase, the existing manual system of vehicle rental booking and payment management is studied carefully. It is observed that manual methods lead to errors in calculation, confusion in booking records, and delays in customer service.

The need for an automated system is identified to simplify booking operations, calculate rental charges accurately, and manage payment details efficiently. The system requirements are defined, including inputs such as vehicle type, rental duration, and payment details, and outputs such as total rental amount and booking summary.

4.2. System Design

In this stage, the overall structure of the system is planned. The application is divided into different modules such as vehicle display, booking, rental calculation, and payment management.

The system flow is designed to ensure that users can easily interact with the system. The design focuses on simplicity, clarity, and efficiency so that the application can be used without difficulty.

4.3. Program Development

In this phase, the actual coding of the system is carried out using the Java programming language. Core Java concepts such as classes, objects, arrays, loops, and conditional statements are used to implement the logic.

The program includes features such as displaying available vehicles, accepting booking details, calculating rental charges, processing payments, and generating booking summaries. Proper coding practices are followed to ensure readability and accuracy.

4.4. Testing and Validation

Edit with WPS Office

Testing is performed to ensure that the system works correctly and produces accurate results. Different test cases are used to verify the functionality of the system.

Example test cases include:

- Selecting different types of vehicles
- Entering various rental durations
- Checking correct calculation of rental charges
- Verifying payment processing
- Ensuring proper display of booking summary

Any errors identified during testing are corrected to improve system performance and reliability.

4.5. Result and Documentation

After successful testing, the system results are verified and documented. The system is able to perform vehicle booking and payment calculation accurately based on user input.

The final output includes booking details, rental amount, payment information, and summary. All stages of the project such as analysis, design, development, and results are documented clearly for academic and practical reference.

Edit with WPS Office

4.6. Algorithm

Start

- Display available vehicles with rental charges
- Ask the user to select a vehicle
- Enter customer details
- Enter rental duration
- Calculate rental amount
- Display total amount
- Accept payment details
- Display booking summary
- End

4.7. Flowchart

Start
↓
Display Vehicles
↓
Select Vehicle
↓
Enter Details
↓
Calculate Rent
↓
Process Payment
↓
Display Summary
↓
End

Students can draw using:

- draw.io
- Lucidchart

Edit with WPS Office

- MS Visio

5. SOURCE CODE

```
import java.util.Scanner;

public class VehicleRentalSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String[] vehicles = {"Bike", "Car", "Scooter", "Van"};
        double[] rentPerDay = {500, 1500, 400, 2500};
```


Edit with WPS Office

```
        System.out.println("===== Vehicle Rental Booking and Payment
Management System =====");
        System.out.println("\nAvailable Vehicles:");
        for (int i = 0; i < vehicles.length; i++) {
            System.out.println((i + 1) + ". " + vehicles[i] + " - Rs." + rentPerDay[i] + "
per day");
        }

        System.out.print("\nEnter customer name: ");
        sc.nextLine();
        String customerName = sc.nextLine();

        System.out.print("Select vehicle (1-4): ");
        int choice = sc.nextInt();

        if (choice < 1 || choice > 4) {
            System.out.println("Invalid vehicle selection.");
            return;
        }

        System.out.print("Enter number of rental days: ");
        int days = sc.nextInt();

        double totalAmount = rentPerDay[choice - 1] * days;

        System.out.println("\n===== Booking Details =====");
        System.out.println("Customer Name  : " + customerName);
```


Edit with WPS Office

```
System.out.println("Vehicle Type   : " + vehicles[choice - 1]);
System.out.println("Rental Days    : " + days);
System.out.println("Rent Per Day   : Rs." + rentPerDay[choice - 1]);
System.out.println("Total Amount   : Rs." + totalAmount);

System.out.print("\nEnter payment amount: Rs.");
double payment = sc.nextDouble();

System.out.println("\n===== Payment Details =====");
if (payment == totalAmount) {
    System.out.println("Payment Status : Paid in Full");
    System.out.println("Balance        : Rs.0");
} else if (payment > totalAmount) {
    System.out.println("Payment Status : Paid in Full");
    System.out.println("Return Amount  : Rs." + (payment - totalAmount));
} else {
    System.out.println("Payment Status : Partial Payment");
    System.out.println("Balance Due    : Rs." + (totalAmount - payment));
}

System.out.println("\n===== Final Booking Summary =====");
System.out.println("Customer Name  : " + customerName);
System.out.println("Vehicle Booked : " + vehicles[choice - 1]);
System.out.println("Rental Days    : " + days);
System.out.println("Total Amount   : Rs." + totalAmount);
System.out.println("Paid Amount    : Rs." + payment);
```


Edit with WPS Office

```
        System.out.println("\nThank you for using Vehicle Rental Booking System!");

        sc.close();
    }
}
```

6. INPUT AND OUTPUT SCREENS

```
===== Vehicle Rental Booking and Payment Management System =====

Available Vehicles:
1. Bike - Rs.500.0 per day
2. Car - Rs.1500.0 per day
3. Scooter - Rs.400.0 per day
```

Edit with WPS Office

4. Van - Rs.2500.0 per day

Enter customer name: Arun

Select vehicle (1-4): 2

Enter number of rental days: 3

===== Booking Details =====

Customer Name : Arun

Vehicle Type : Car

Rental Days : 3

Rent Per Day : Rs.1500.0

Total Amount : Rs.4500.0

Enter payment amount: Rs.5000

===== Payment Details =====

Payment Status : Paid in Full

Return Amount : Rs.500.0

===== Final Booking Summary =====

Customer Name : Arun

Vehicle Booked : Car

Rental Days : 3

Total Amount : Rs.4500.0

Paid Amount : Rs.5000.0

Thank you for using Vehicle Rental Booking System!


Edit with WPS Office

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system was successfully implemented using the Java programming language, utilizing core concepts such as classes, objects, loops, and conditional statements. It is designed to perform GPA and CGPA calculations accurately by following standard mathematical formulas, ensuring reliable and consistent results.

Edit with WPS Office

The application effectively reduces manual effort by automating the entire calculation process, which minimizes the chances of human errors and saves valuable time for students and academic staff. It provides a simple and user-friendly interface, making it easy for users to input data and obtain results quickly.

Extensive testing was carried out using different input values, including varying numbers of subjects, credits, and grade points. The outputs generated by the system were compared with manual calculations and were found to be correct and precise. This confirms the accuracy and dependability of the system.

7.2 Conclusion

The Semester GPA and CGPA Calculation System is a simple yet highly effective solution designed to assist in the accurate calculation of academic performance. By automating the process of GPA and CGPA computation, the system eliminates the need for manual calculations, thereby significantly reducing the chances of human errors. It also saves time and effort for students and academic staff, making the evaluation process faster and more efficient.

The system is developed using Java programming and demonstrates the practical application of core concepts such as loops, conditional statements, and user input handling. It provides a user-friendly interface that allows users to easily enter subject details, credits, and grades to obtain instant results. This makes the system convenient and accessible even for users with basic technical knowledge.

In addition, the system ensures consistency and accuracy in results by following standard calculation methods. It can handle multiple subjects and semesters, making it suitable for real academic scenarios. The project also helps students understand their performance clearly and track their academic progress effectively.

8. REFERENCES

Herbert Schildt – Java: The Complete Reference

- www.oracle.com
- www.geeksforgeeks.org

Edit with WPS Office

- www.w3schools.com

Edit with WPS Office